

Optimizing Agricultural Resource Allocation through Reinforcement Learning: A Cost-Driven Approach to Crop Efficiency Enhancement

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OCTOBER 2025

PRESENTATION OUTLINE

- Introduction
- Motivation
- Literature Review
- Research Gaps
- Problem Statement
- Solution Statement
- Research Objectives
- Approach /
Methodology
- Proposed Timeline

INTRODUCTION

INTRODUCTION

PAKISTAN AN AGRICULTURAL NATION

- ~22% of GDP.
- Major irrigated area
- Exports share (rice, cotton)
- Employs 36.1% of labor force

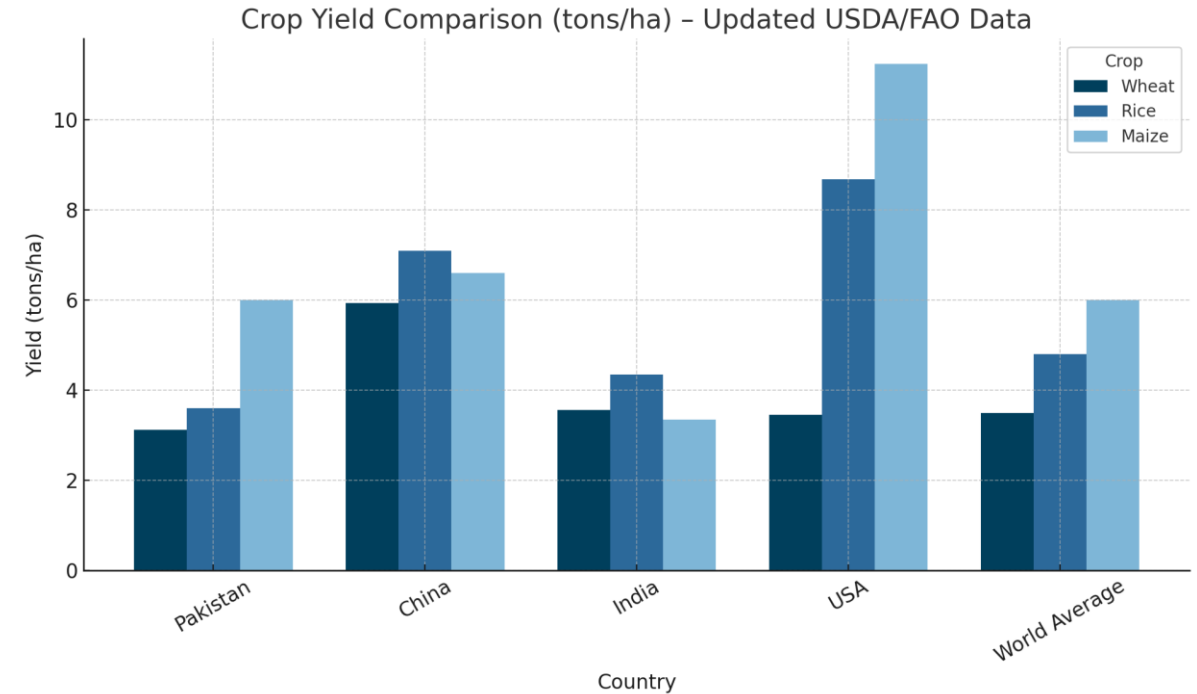


INTRODUCTION

Global Agricultural Production Comparison

Pakistan's Agricultural Performance vs. Global Standards

Country	Wheat Yield (tons/ha)	Rice Yield (tons/ha)	Maize Yield (tons/ha)
Pakistan	2.8	3.6	4.9
China	5.4	7	6.3
India	3.5	4	3.2
USA	3.2	8.7	11.1
World Average	3.4	4.6	5.9

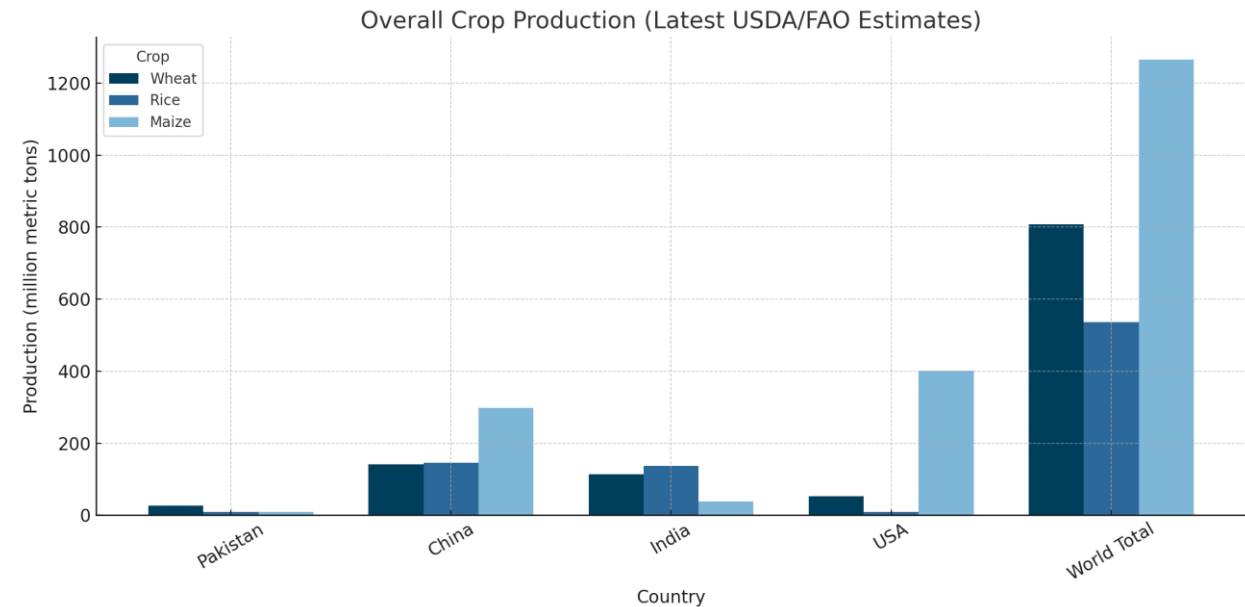


INTRODUCTION

Global Agricultural Production Comparison

Pakistan's Agricultural Performance vs. Global Standards

Country	Wheat	Rice (milled-eq.)	Maize
Pakistan	27.5 MMT	9.8 MMT	9.6 MMT
China	141 MMT	146 MMT	298 MMT
India	113.3 MMT	137.8 MMT	37.7 MMT
United States	≈52 MMT (1.921 bn bu) (downloads.usda.library.cornell.edu)	≈10 MMT (222 M cwt) (isuagcenter.com)	≈402 MMT (15.82 bn bu) (feedandgrain.com)
World total	808.5 MMT (graincentral.com)	535.8 MMT (ers.usda.gov)	1 265 MMT (usda.gov)



INTRODUCTION

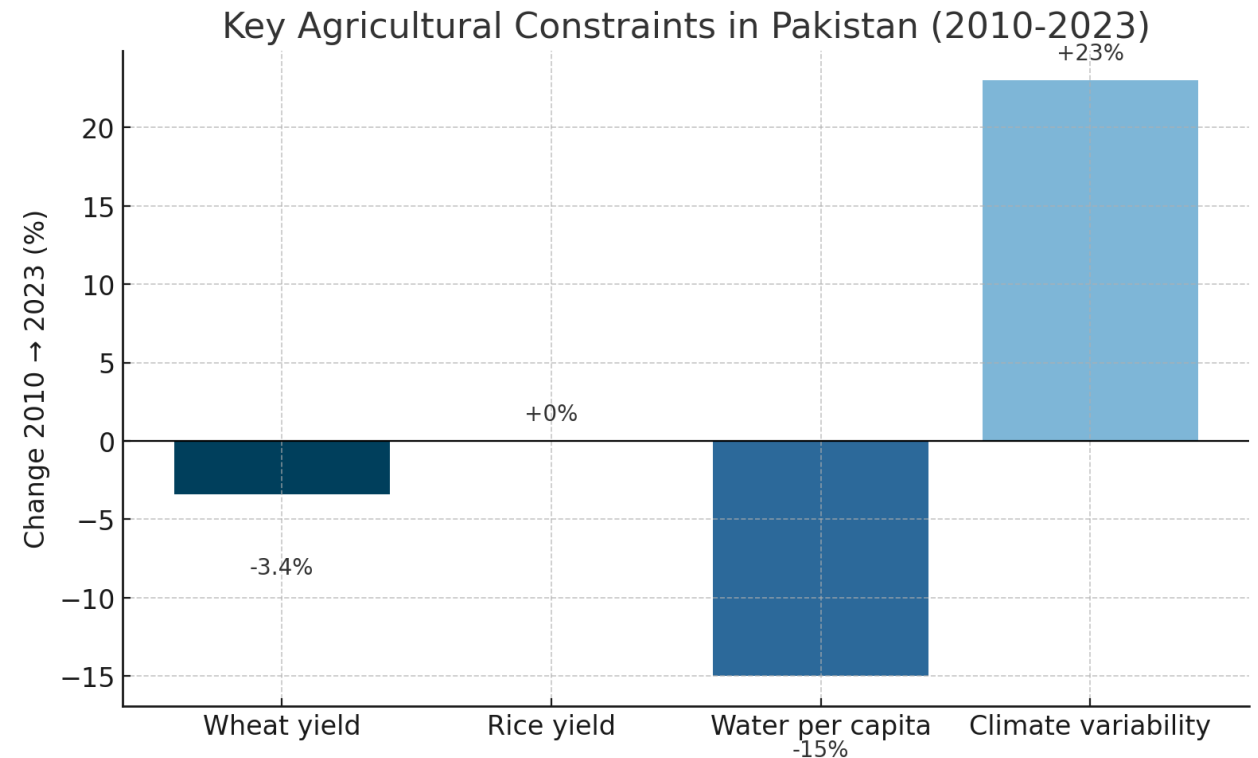
Productivity Headwinds (2010-2023)

Key facts:

- Wheat yield ↓ **3 %** (2.9 → 2.8 t ha⁻¹)
- Rice yield **flat** (≈ 3.6 t ha⁻¹)
- Per-capita water ↓ **15 %**
- Climate-variability index ↑ **23 %**

Reasons:

- Inefficient fertilizer timing
- Flood-irrigation losses > 40 %
- Low adoption of precision tech
- More heatwaves, floods, droughts



MOTIVATION

MOTIVATION

Comparison of Manual Farming Practice with Automated or Smart Farming Practice

Characteristic	Manual Farming	Automated Farming
 Decision Making	Subjective	Objective
 Response Time	24–48 h lag	Near-real-time
 Data Handling	Data used in silos	Data integrated
 Fertiliser Variation	30–40 %	5 %
 Waste	~25 %	10–15 %
 Scalability	Hard to scale	Easily replicable



LITERATURE REVIEW

LITERATURE REVIEW

Machine Learning Approaches in agriculture and their limitations

Paper Title	Authors	Strengths	Weaknesses	Dataset Used	Venue & Year
Integrating Random Forest & Crop Modeling Improves Yield of Winter Wheat & Oilseed Rape [1]	Dhillon et al.	- Hybrid RF + LUE model lifts R^2 & cuts RMSE vs. RF w/o crop-model inputs. - Uses NDVI + climate $\leftrightarrow$ crop growth priors.	- Tested in one European region; transferability unknown. - Needs crop-model calibration.	Sentinel-2 NDVI, ERA5 climate; field yields (Germany)	<i>Frontiers in Remote Sensing</i> , 2022
CLIMATES: Mitigating Low Productivity on African Smallholder Farms [2]	Tabar et al.	- Clustering + per-cluster ensemble lowers error vs. LSTM/VRNN. - Handles 2.2 k farm time-series with diverse climates.	- Relies on quality cluster labels. - Designed for environmental stress forecasting, not prescriptions.	Environmental time-series (AET, RET, NPP) from 2 200 African farms	AAAI 2022

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Paper Title	Authors	Strengths	Weaknesses	Dataset Used	Venue & Year
Optimal Greenhouse Sensor Placement via GBM Grid-Search [3]	Uyeh et al.	- Gradient Boosting selects minimal sensors with lowest RMSE. - Psychrometric feature engineering improves predictions.	- Focus on temperature/humidity only. - Validation limited to two greenhouses.	IoT sensor readings (T, RH) in protected cultivation	<i>Frontiers in Plant Science</i> , 2022
Explainable Crop Recommendation with Ensemble ML [4]	Shastri et al.	- Compares 10 algorithms; Gradient Boosting highest accuracy. - LIME explains nutrient/weather influence on crop choice.	- Static recommendation (no temporal dynamics). - Tested on Indian soils only.	Soil N-P-K & weather data (India)	<i>Scientific Reports</i> , 2025

LITERATURE REVIEW

Deep Learning Approaches in agriculture and their limitations

Paper Title	Authors	Strengths	Weaknesses	Dataset Used	Venue & Year
CropAndWeed Dataset: Multi-Modal Learning for Precision Weeding [5]	Steininger et al.	8 k aerial RGB images → 112 k instances. - CNN detector benefits from species diversity; supports detection + segmentation.	Drone imagery only; ground-level viewpoint not tested.	CropAndWeed (74 species, images + meta)	WACV 2023
iGrow: Autonomous Greenhouse Control with Deep RL [6]	Cao et al.	Neural “digital twin” + deep RL → +10.2 % tomato yield, +92.7 % profit vs. experts.	- Domain-specific simulator requires calibration per greenhouse. - RL generalisation across climates untested.	Real tomato greenhouse sensor/control logs	AAAI 2022

LITERATURE REVIEW

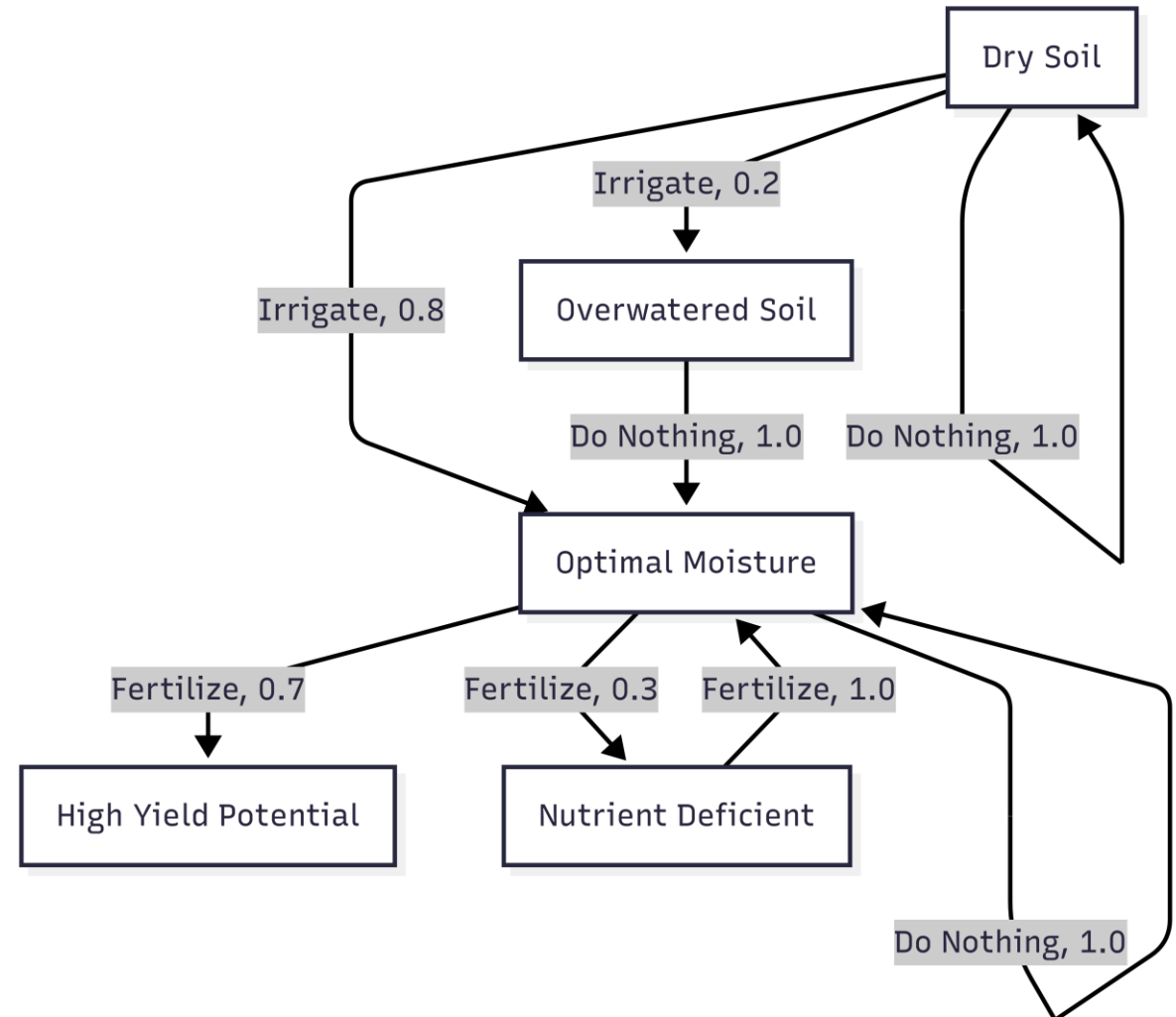
Deep Learning Approaches in agriculture and their limitations

Paper Title	Authors	Strengths	Weaknesses	Dataset Used	Venue & Year
Deep Diagnosis: Real-time Apple Leaf Disease Detection [8]	Khan et al.	Lightweight CNN runs in real-time on edge devices; high accuracy on multiple apple diseases.	Apple-specific; cross-crop transferability untested.	Custom apple leaf image set (lab + orchard)	<i>Computers & Electronics in Agriculture</i> , 2022
Augmentation Invariance & Adaptive Sampling for Aerial Segmentation [9]	Tavera et al.	- Consistency loss + adaptive sampling $\uparrow$ mIoU on Agriculture-Vision. - Robust to rotations/lighting changes.	- Benefits diminish on ground imagery. - Adds training complexity.	Agriculture-Vision (aerial RGB-NIR)	CVPR Workshops 2022
Utilizing Vision-Language Models for Detection of Leaf-Based Diseases in Tomatoes [10]	James B. Eleojo	- Leverages a pre-trained vision-language model $\rightarrow$ strong few-shot generalisation - Achieved F1 $\approx$ 0.94 on field-captured leaves	- Compute-heavy (VLM fine-tuning) - Trained on tomato only $\rightarrow$ limited crop transferability	12 k labelled tomato-leaf images (greenhouse + field)	AAAI 2025 (AI for Social Impact)

LITERATURE REVIEW

Why Reinforcement Learning for Agriculture ?

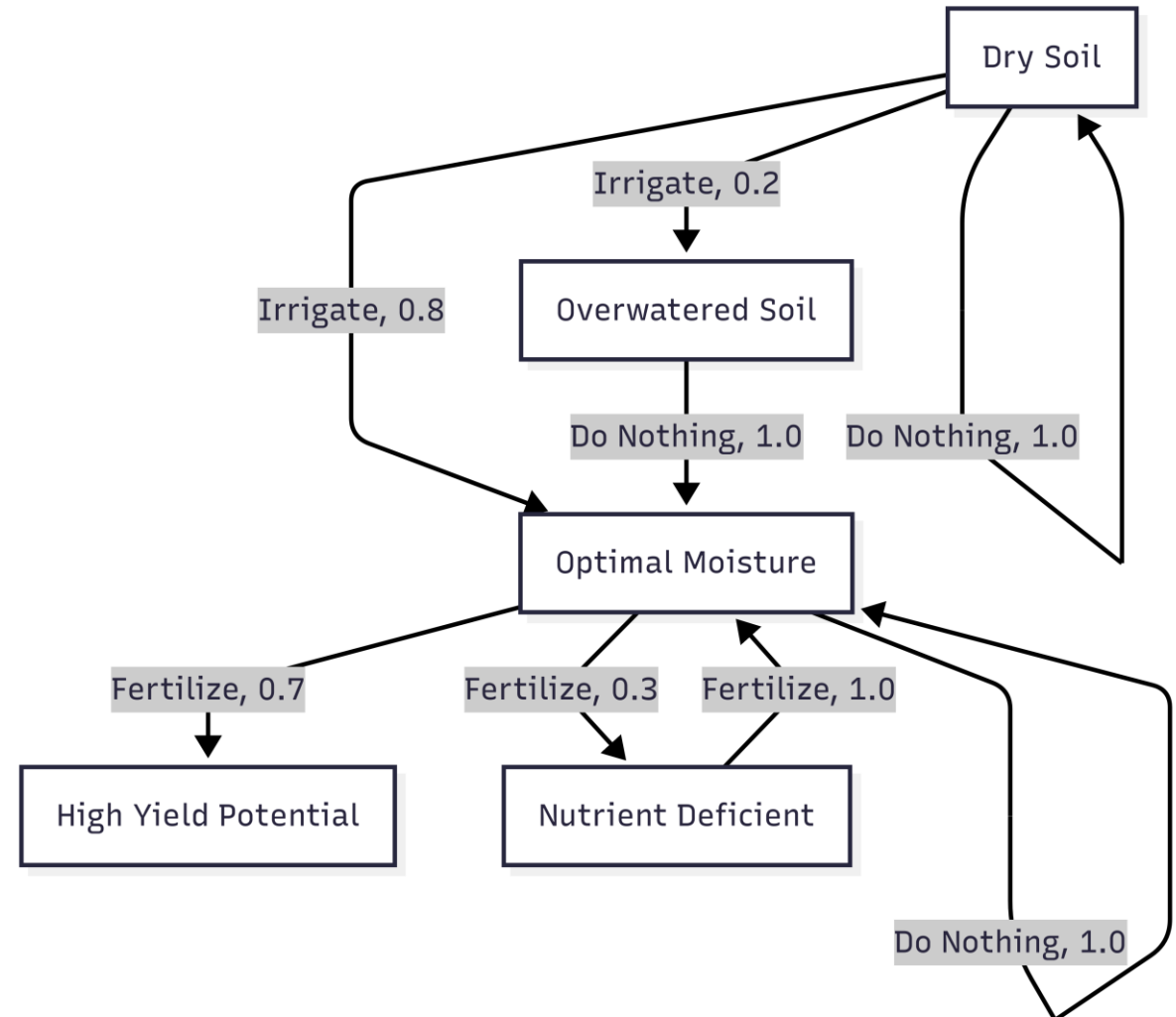
- **Interactive learning** – agent explores crop/soil climate environment through trial-and-error and continuously improves its policy
- **Sequential decision-making (MDP)** – models season-long input choices as a Markov Decision Process to maximize long-term reward (yield – cost)
- **Real-time adaptation** – dynamically adjusts fertilizer, irrigation, and sowing under weather and pest uncertainty using feedback



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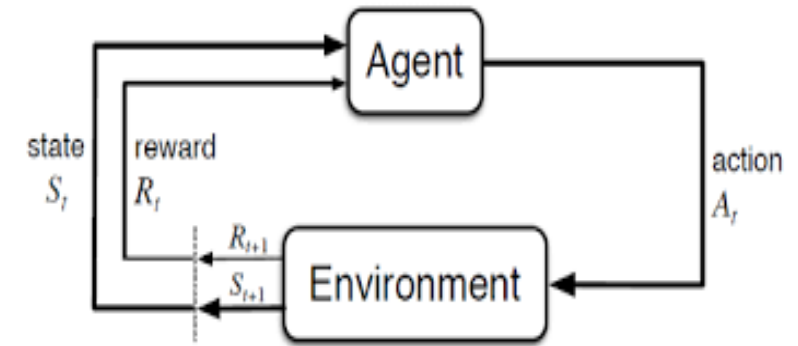
LITERATURE REVIEW

Why Reinforcement Learning for Agriculture ?

Reinforcement learning loop:

The agent interacts with the environment, receiving observations and rewards.

- **AN AGENT** (think of it as an AI farmer) observes the environment (the farm simulation) and decides on an action (e.g. how much to irrigate or fertilize).
- The **ENVIRONMENT** then updates the crop growth and returns a reward (a score) and a new state back to the agent.
- This **LOOP REPEATS OVER TIME**, and the agent's goal is to learn a policy that maximizes the cumulative reward (for example, high crop yield minus input costs)



LITERATURE REVIEW

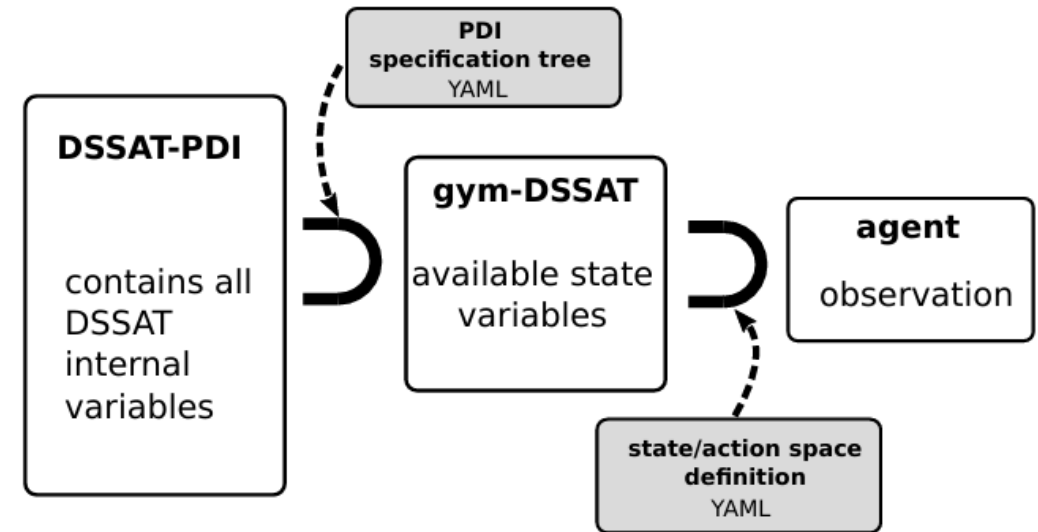
RL PAPER 1 : CropGym: a Reinforcement Learning Environment for Crop Management (Environmental Data Science by Cambridge University Press)

- **Model Foundation:** LINTUL-3 winter wheat module (light-use efficiency based)
- State Representation (18-dimensional)
 - Development stage, Leaf area index, Total biomass (kg/ha), Available soil nitrogen
 - Environmental Context : Temperature, Rainfall, Solar radiation, 5-day aggregated summaries
 - Management Tracking : Recent Fertilization/ Irrigation Flags
- Action Space : Agent applies nitrogen fertilizer at 5-day intervals based on observed state
- Optimization Framework
 - $$r_t = \frac{m_{so,t} - m_{so,t-1}}{m_{so,t}^* - m_{so,t-1}^*} - \beta m_{fert,t}$$
 - The reward balances crop yield improvement against fertilizer cost, where β controls the economic-environmental trade-off ($\beta=1$ for profit, $\beta>1$ for sustainability).

LITERATURE REVIEW

RL PAPER 2 : Gym-DSSAT: a crop model turned into a Reinforcement Learning environment (Research Report N° 9460 by Inria)

- **Crop Model:** DSSAT (Decision Support System for Agrotechnology Transfer)
- **Implementation:** Fortran-Python coupling via PDI
- **Crops:** Maize (extensible to 41+ crops)
- **Episode:** Full growing season (~160 days)
Decisions: Daily time steps
Weather: WGEN stochastic generator
Termination: Crop maturity or early failure
- **State Space (13 variables)**
- Growth stages, biomass, nitrogen stress, leaf area, weather, cumulative inputs



Reward Functions

Fertilization:

$$r(t) = \text{trnu}(t, t + 1) - 0.5 \times \text{anfer}(t)$$

Irrigation:

$$r(t) = \Delta \text{biomass}(t) - \text{penalty} \times \text{irrigation}(t)$$

LITERATURE REVIEW

Why Long-Term Planning in Agriculture Needs AI

Modern agriculture faces a multi-objective challenge:

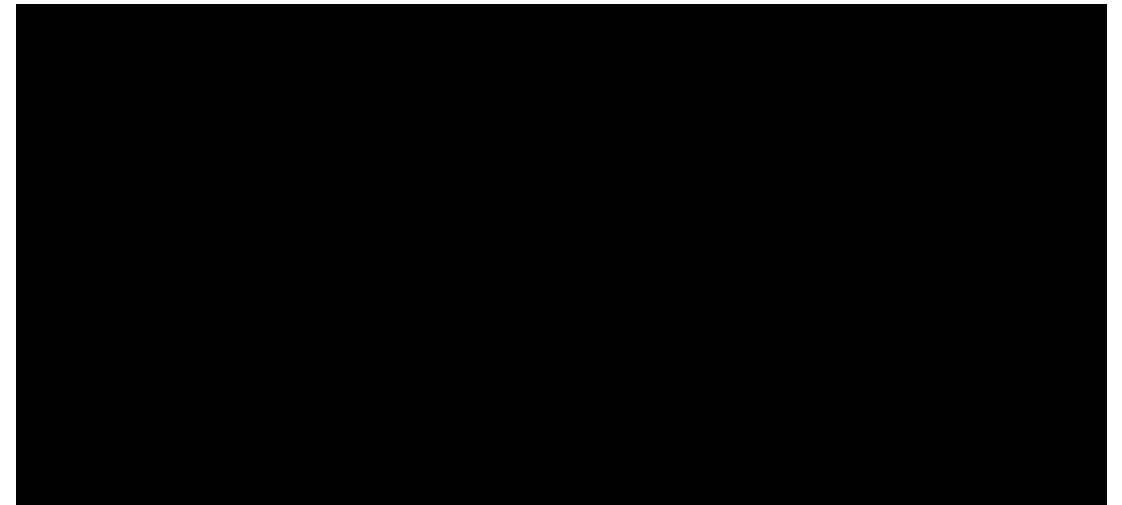
- Maximize profit while minimizing environmental harm and adapting to weather variability and soil degradation.

Farmers often make single-season decisions, but in reality, agriculture is interdependent across years:

- Fertilizer applied this year affects soil nitrogen next year
- Crop choices affect pests, diseases, and water balance
- Unplanned fertilizer use causes waste and pollution

What is Leaching?

- When **too much nitrogen** fertilizer is applied, the excess **leaches** — it dissolves in rainwater and **moves beyond the root zone**



LITERATURE REVIEW

RL PAPER 3 : Long-Term Crop Management Strategies with CyclesGym (NeurIPS 2022)

	Year 1		Year 2		Reduced Leaching	Reduced Emissions	Profit
Environment							
Management Strategies	Maize	Fertilization	Maize	Fertilization	✗	✗	✓
	Soy		Soy		✓	✓	✗
	Soy		Maize	Low Fertilization	✓	✓	✓

Strategy	Profit	Environmental impact	Why / mechanism
Monoculture maize + non-organic fertilization	High ↑	High ↑	Heavy fertilizer → big yields, but N leaching + GHG emissions rise
Soybean focus with low fertilization	Lower ↓	Low ↓	Lower yields from soy; minimal external N input
Maize–soybean rotation , leverages soy N-fixation	High ↑	Medium–Low ↘	Soy adds N biologically → less fertilizer for maize, decent yields

LITERATURE REVIEW

RL PAPER 3 : Long-Term Crop Management Strategies with CyclesGym (NeurIPS 2022)

Multi-Year Planning:

$$V^{\pi}(s) = \mathbb{E}^{\pi} \left[\sum_{t=0}^T \gamma^t r_t \middle| s_0 = s \right]$$

where T spans multiple growing seasons

Crop Rotation Benefits:

- Nitrogen fixation modeling:

$$N_{available,t+1} = N_{soil,t} + N_{fixed,t} + N_{fertilizer,t}$$

- Soil health improvements

Constraint Integration:

- Environmental limits: $\text{Leaching}_t \leq \text{threshold}$
- Economic constraints: $\sum_t \text{costs}_t \leq \text{budget}$
- Regulatory compliance modeling

Complex Action Spaces:

- Timing decisions: $a_t = (\text{crop_type}, \text{plant_date}, \text{fertilizer_amount}, \text{timing})$
- Multi-dimensional optimization
- Seasonal planning integration

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RESEARCH GAP

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Regional Agricultural Challenges

- **Reinforcement Learning Based Approaches in Pakistan**
 - RL has been applied in greenhouse environments or precision farming in developed countries, but there is almost no work in open-field, smallholder contexts in South Asia.
- **Absence of Multi-Objective Optimization:**
 - Existing models often optimize only yield or revenue, not long-term soil health, water sustainability, or resilience to climate variability.
- **Climate Adaptation:**
 - No adaptation to shifting climate patterns
- **No Integration with Local Economic Constraints:**
 - Most existing models do not account for smallholder farm sizes, limited capital, and fluctuating input prices common in Pakistan.
- **Data-to-Decision Gap:**
 - Research largely remains in academic papers without being translated into tools for local farmers or policymakers.

PROBLEM STATEMENT

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“ Existing reinforcement learning-based agricultural management systems are developed for North American and European contexts and cannot be directly applied to Pakistani agricultural conditions due to different climate patterns, crop varieties, farming practices, and economic constraints ”

SOLUTION STATEMENT

SOLUTION STATEMENT

Proposed Solution: Pakistan-Adapted Agricultural RL Framework

Core Solution:

Develop a comprehensive reinforcement learning framework based on CyclesGym architecture, adapted for Pakistani agricultural context with integrated climate modeling, local crop varieties, and economic constraints.

Key Components:

- 1. Climate Module:** Monsoon pattern integration and drought modeling
- 2. Crop Adaptation:** Pakistani wheat, rice, cotton, and maize varieties
- 3. Soil Data:** Local soil data

RESEARCH OBJECTIVES

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- **Develop a CyclesGym-PK simulation environment**
 - Inject monsoon weather series, Indus-plain soil profiles and local wheat/rice cultivars
- **Design a cost-aware reward function**
 - Maximise net profit while penalising water use and N-leaching
- **Train a PPO agent for weekly irrigation scheduling**
 - Learn policies under stochastic monsoon rainfall data

RESEARCH METHODOLOGY

RESEARCH METHODOLOGY

Methodology Overview

Data stream	Source & format
Weather (30 y)	Pakistan Meteorological Dept. → <i>.weather</i>
Soil profile	FAO Indus-loam → <i>.soil</i>
Crop cultivar	IRRI-6 rice, Sehar-06 wheat → <i>GenericCrops.crop</i>
Price & cost constants	Pakistan Bureau of Statistics market bulletins → <i>reward_config.json</i>

- Weather generator: PKWeatherGenerator(csv_dir, delta_days=7)
- Soil generator: StaticSoilGenerator("IndusLoam.soil")
- Rotation file: rice → wheat 2-year .operation template (monsoon sow 10 Jul; rabi sow 15 Nov)
- Delta & horizon: delta_days=7, duration_years=2
- Baseline check: Simulator reproduces $\approx 3.6 \text{ t ha}^{-1}$ rice & 2.8 t ha^{-1} wheat with farmer schedule

RESEARCH METHODOLOGY

PPO Agent Setup

Setting	Value	Rationale
NN	2 × 64 dense (Tanh)	Standard PPO baseline
Actions	irrigation $\in \{0, 25, 50 \text{ mm week}^{-1}\}$	Keeps search space small
Batch	16 environments × 128 steps	$\approx 2\,500 \text{ wk batch} \approx 5 \text{ CPU-min}$
Hyper-params	γ : 0.99 Clip: 0.2 Learning Rate: $3\text{e-}4$ Entropy: 0.01	Default stable config
Training budget	100 k episodes ($\approx 8 \text{ CPU-hrs}$)	



RESEARCH METHODOLOGY

Reward Shaping Matters

Reward shaping is not just tuning numbers it's defining the problem itself.

Why it's essential in CyclesGym:

- Pakistani agriculture has unique economics (crop prices, water costs, fertilizer subsidies).
- Environmental penalties (e.g. nitrate limits) are region-specific and policy-relevant.
- Without proper shaping, the agent may learn:
 - Costly over-irrigation,
 - Excess fertilization (high leaching),
 - Or exploit unrealistic dynamics.

Why Reward Shaping is Non-Trivial:

- Even small changes (e.g., soil type, cultivar) alter how actions impact future rewards → it becomes a new MDP, not just a new dataset.
- RL agents must re-learn the effect of each action in this changed world transferring old policies fails.
- Penalty weights (β) require careful tuning to balance economics vs. sustainability.
- It encodes local reality into the simulator and ensures the RL agent's learned behavior is aligned with real-world goals.

RESEARCH METHODOLOGY

Evaluation & Robustness Plan

1. **Weather sets:** dry (2019), median (2020), wet (2022)
2. **Soil PAW ± 10 %:** test shallow vs. deep loam
3. **Price shock:** urea + 15 %, rice price – 10 %

KPIs

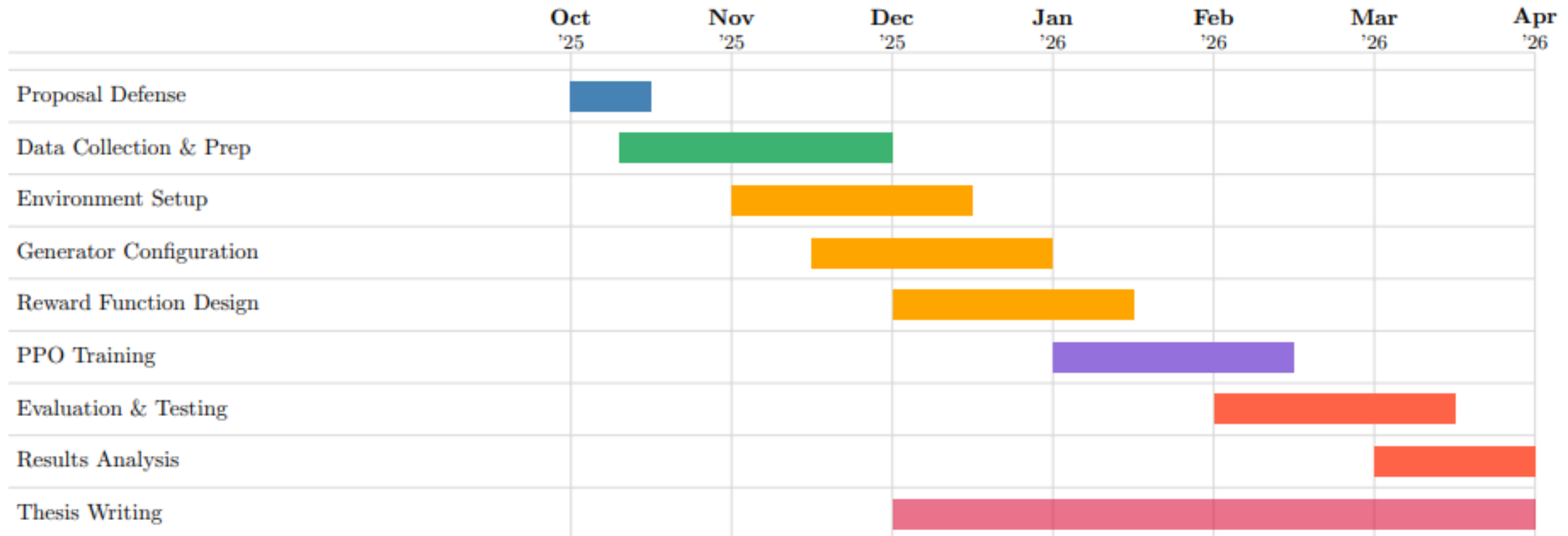
- Yield (t ha^{-1})
- Irrigation (mm)
- Net profit (PKR ha^{-1}) – **computed by reward class**
- N-use efficiency (%)
- Constraint breaches ($\text{N-leach} > 40 \text{ kg ha}^{-1} \text{ yr}^{-1}$)



TIMELINE

Timeline

October 2025 – April 2026



CONCLUSION

Conclusion

Key Contributions:

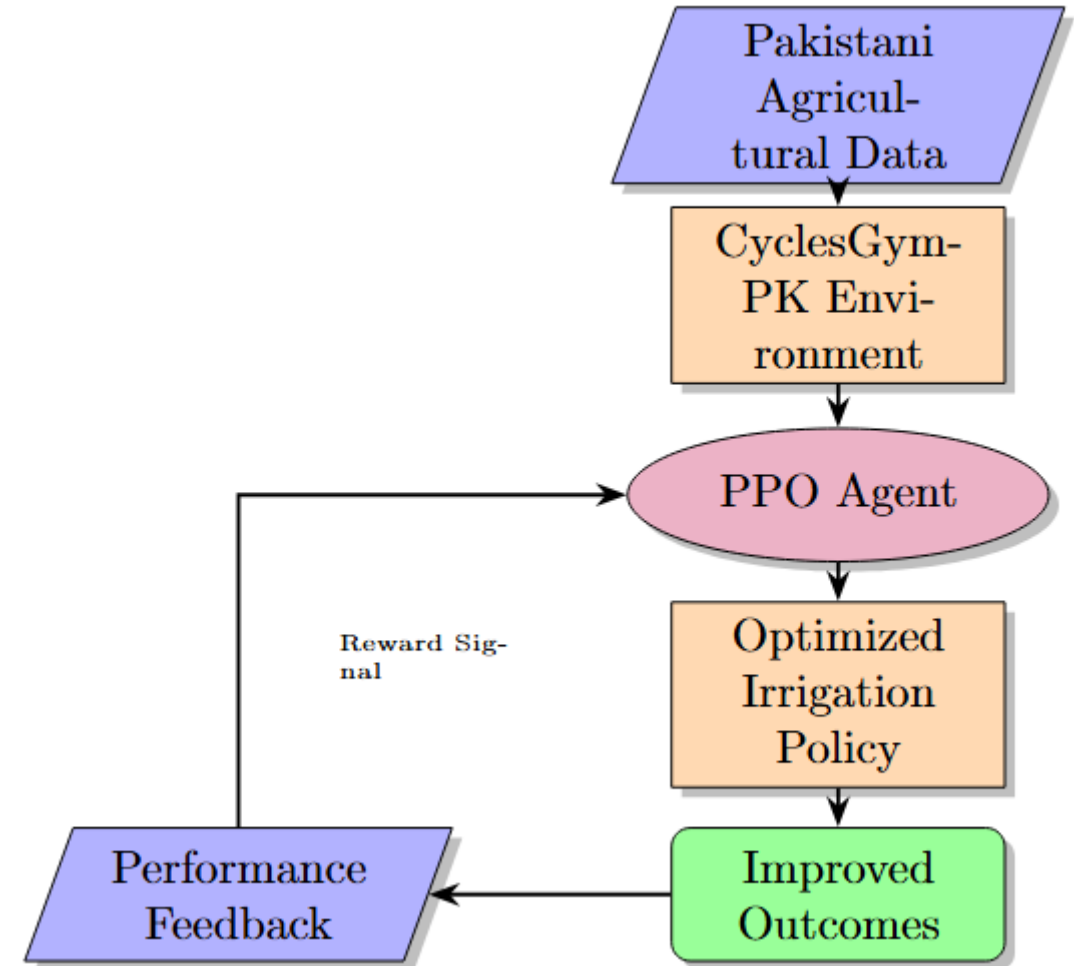
- First RL-based agricultural framework adapted for Pakistani context
- Integration of local climate patterns, soil data, and economic constraints
- Cost-aware reward function balancing profit and sustainability
- Practical irrigation scheduling tool for smallholder farms

Expected Impact:

- Reduced nitrogen leaching and environmental footprint
- Data-driven decision support for farmers
- Extensible framework for other South Asian regions

Future Directions:

- Multi-crop rotation optimization
- Real-world pilot deployment with farmers
- Integration with IoT sensors and mobile platforms



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